

University Saleh Boubnider Constantine 3
Laboratory of General Anatomy.
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MUSCLES OF THE THIGH

PLAN

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I. INTRODUCTION:

The muscles of the thigh are divided into three groups:

- An anterior group: extensor muscles.
- A medial group: adductor muscles.
- A posterior group: flexor muscles.

The thigh is enveloped by a superficial fascia and a deep fascia (fascia lata).

II. DESCRIPTIVE ANATOMY:

A. **ANTERIOR MUSCLE GROUP:** Includes 3 muscles arranged in two layers.

❖ Deep Layer

1. **Quadriceps femoris muscle** composed of:

- Vastus intermedius
- Vastus medialis
- Vastus lateralis
- Rectus femoris

a) **Vastus Intermedius:**

- Origin: Anterior and posterolateral surfaces of the femur.
- Body: thick, curved, gutter-shaped muscle.
- Insertion: Quadriceps tendon.
- Action: Extension of the leg.
- Innervation: Femoral nerve (lumbar plexus).

b) **Vastus Medialis:**

- Origin: Medial lip of the linea aspera and spiral line.
- Body: broad and thick muscle blade.
- Insertion: Quadriceps tendon.
- Action: Extension of the leg.
- Innervation: Femoral nerve (lumbar plexus).

c) **Vastus Lateralis:**

- Origin: Greater trochanter (anterior and inferior border) and lateral lip of the linea aspera.
- Body: broad and thick muscle blade.
- Insertion: Quadriceps tendon.
- Action: Extension of the leg.
- Innervation: Femoral nerve (lumbar plexus).

d) **Rectus Femoris:**

- Origin:
 - Direct tendon: on the Anterior inferior iliac spine
 - Reflexed tendon: supra-acetabular groove
 - Recurrent tendon: intertrochanteric line.
- Body: long, flattened and superficial muscle.
- Insertion: Base of the patella via quadriceps tendon.
- Action: Extends the leg and flexes the thigh at the hip.
- Innervation: Femoral nerve (lumbar plexus).

2. **Articularis Genu muscle:**

- Origin: Anterior surface of femur below vastus intermedius.
- Body: fleshy bundle attached to the vastus intermedius.
- Insertion: Suprapatellar bursa.
- Action: Elevates the synovial bursa.
- Innervation: Femoral nerve.

❖ Superficial Layer:

3. **Sartorius muscle :**

- Origin: Anterior superior iliac spine.
- Body: very long flattened muscle.
- Insertion: Upper medial surface of the tibia.
- Action: Flexes the leg and thigh.
- Innervation: Femoral nerve.

B. **MEDIAL MUSCLE GROUP:**

❖ Deep Layer:

1. **Adductor Magnus muscle:**

- Origin: Ischiopubic ramus and ischial tuberosity.

- Body: a large, thick and triangular muscle, presenting three bundles: superior, middle and inferior
- Insertion:
 - Upper bundle: on the lateral trifurcation branch of the linea aspera.
 - Middle bundle: in the interstice of the linea aspera.
 - Lower bundle: on the adductor tubercle
- Action: Adduction and lateral rotation of thigh.
- Innervation: Obturator nerve (lumbar plexus) and sciatic nerve (sacral plexus).

❖ Middle Layer:

2. Adductor Brevis muscle:

- Origin: Quadrilateral plate of the pubis and the upper part of the ischiopubic ramus.
- Body: flattened and triangular muscle.
- Insertion:
 - Upper bundle: on the crest of the adductor brevis
 - Lower bundle: in the interstice of the linea aspera.
- Action: Adduction and flexion of thigh.
- Innervation: Obturator nerve

❖ Superficial Layer:

3. Adductor Longus muscle :

- Origin: on the angle of the pubis and the underside of the pubic tubercle..
- Body: flattened and triangular muscle.
- Insertion: in the interstice of the Linea aspera.
- Action: Adduction and flexion of thigh.
- Innervation: Obturator nerve

4. Pectineus muscle :

- Origin: Pecten pubis and the anterior lip of the obturator groove.
- Body: Flattened, quadrilateral muscle.
- Insertion: Pectineal line of femur.
- Action: Adduction and flexion of thigh.
- Innervation: Femoral nerve (sometimes obturator nerve).

5. Gracilis muscle :

- Origin: Pubis near symphysis.
- Body: flattened , ribbon-like muscle
- Insertion: Upper medial tibia (pes anserinus).
- Action: Flexion and medial rotation of leg, adduction of thigh.
- Innervation: Obturator nerve.

C. POSTERIOR MUSCLE GROUP:

1. Semimembranosus muscle:

- Origin: Ischial tuberosity.
- Body: lean , flat muscle
- Insertion:
 - Direct tendon: Medial tibial condyle(posterior face)
 - Reflexed tendon: Medial tibial condyle(anterior face)
 - Recurrent tendon: the lateral condylar shell and the fabella
- Action: Flexes leg and thigh; medial rotation of leg.
- Innervation: Tibial part of sciatic nerve (sacral plexus).

2. Semitendinosus muscle :

- Origin: Ischial tuberosity.
- Body: fusiform muscle
- Insertion: Upper part of the medial surface of the tibia.
- Action: Flexes leg and thigh; medial rotation of leg.
- Innervation: superior and inferior nerves of the semitendinosus (sacral plexus).

3. Biceps Femoris muscle:

- Origin:
 - Long head :ischial tuberosity
 - Short head: lateral lip of the linea aspera.
- Body: composed of two heads, the short head is inside the semimembranosus, the long head is lateral to the semitendinosus.
- Insertion: Head of fibula.
- Action: Flexes leg, extends thigh, lateral rotation of leg.
- Innervation: Sciatic nerve (sacral plexus).

Note:

Pes anserinus muscles include:

- Sartorius (anterior)
- Gracilis (medial)
- Semitendinosus (posterior)

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